Phytotherapy

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Phytotherapy



The branch of herbal medicine that describes the potentials and limitations of herbal drugs in the treatment of human diseases and should be practiced by physicians or pharmacists with knowledge of phytochemistry and phytopharmacology.

❖ Herbal Drugs: (Phyto-pharmaceuticals = Herbal medicines)



- They are products containing plant material (comminuted or powdered vegetable drugs) or their crude products (extracts, tinctures, fatty or essential oils, plant juices, etc....) used for treatment of disease (usually chronic) or to maintain a condition of improved health.
- It may include also animal, fungi or bacteria products
- They don't include defined isolated constituents or their mixtures.

Herbal medicine

Crude drug vs. pure compound(s) (synthetic or natural) in Modern era

- The Focus is more on the identification and isolation of active ingredients from plants rather than studying their pharmacological and clinical properties.
- Start of the dominance of chemistry by synthesis of medicinal active pure compound(s) of natural origin and their chemical analogues as conventional medicine.
- Nowadays, Herbal medicine using crude drugs is a branch of non-conventional medicine.





Contribution of Natural Products In Global Market

In the raw form (non-conventional medicine)

Overall international trade in medicinal plants and their products was US$ 60 billion in 2010. e.g. waters as sea water, thermal water and rose water.

Pure compounds (natural but conventional medicine)

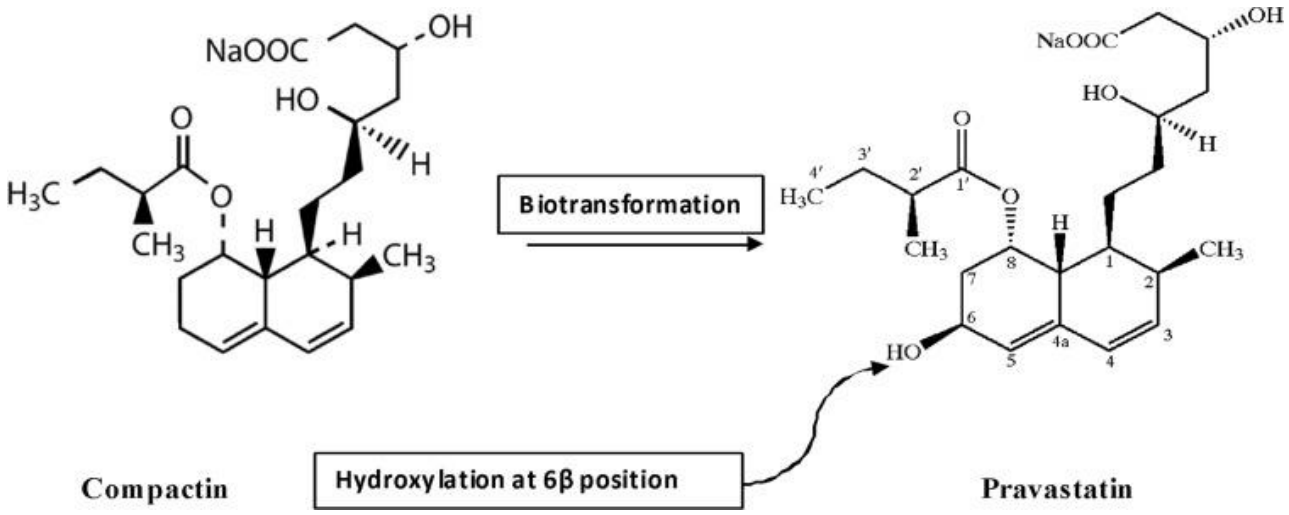
pravastatin (US$3.6 billion per year), is made through different biotransformation processes from compactin by Streptomyces carbophilus and Actinomadura sp.







Combactin to pravstatin



Advantages of the Biotransformation Process

•**Higher specificity**: Microbial hydroxylation selectively introduces a hydroxyl group at C6, which is difficult to achieve by chemical synthesis.

•**Lower toxicity:** Pravastatin is more hydrophilic, reducing hepatic metabolism and minimizing drug interactions.

•**Environmentally friendly:** Microbial transformation is more sustainable than chemical synthesis.

Classification of Herbal drugs

Herbal drugs types acc. to processing:

1- Unprocessed



➤ **Crude Drug:** natural products which are not pure compound as plants, extracts or exudates used as such or unprocessed.

2-Processed in dosage form

A-Phytopharmaceuticals: Is the natural biologically active fraction or standardized extracts formulated in certain pharmaceutical dosage form and legally approved for intended use and user.



Classification of Herbal drugs

- **2-Processed in dosage form**
- **B-Nutraceutical or food supplement:**
- Natural substances isolated and formulated in certain dosage form used as nutrient or aid the body health to prevent or protect from certain disease such anthocyanidins, isoflavones and carotenoids.



Classification of Herbal drugs



I-According to strength of the extract

1- Aqueous extracts: as teas, infusions and decoctions.

2-Hydro-alcoholic extracts



A- Tincture: liquid dosage form prepared by solvent extraction e.g. alcohol 1:5 strong tincture or weak 1:10. (1 gm of dry herb with 5 or 10 ml ethyl alcohol 50-70% by maceration or percolation).

B-liquid extract: liquid dosage form prepared by solvent extraction 1:1 or 1:2 ratio



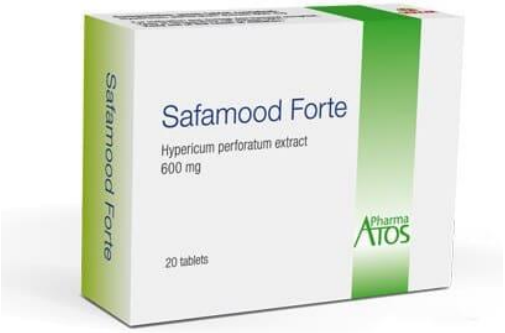
3- dry extract: removing extraction solvent without any residual ex. ginkgo 50:1= 50 gm of dried leaves give 1 gm extract.

Classification of Herbal drugs

II-According to standardization

1-Standardized extract:

this is extract with relatively well-established chemical, pharmacological and clinical profiles.



2-Non-standardized extract:

extract with little information about quality, chemical composition and efficacy.

Classification of Herbal drugs

III-According to potency

1- Highly potent

a- Completely toxic: Not used at all

e.g. Aconite and Nux vomica

b-Contain useful but toxic constituents

Should be prescribed by specialist e.g. Cardiac glycosides containing herbs as Digitalis, Strophanthus and Squill

c-Drastic purgatives

May cause death due to loss of water and electrolytes (dehydration) e.g. Colocynth, Jalap and Podophyllum

Classification of Herbal drugs

2- Intermediately potent:

Should be (preferably) used only as standardized dosage forms e.g. Solanceous herbs, Ephedra and Colchicum

3- Gentle or mild:

Highly safe, non toxic and suitable for self-treatment; the majority of herbal drugs

Total extract vs single compound

> Differences between Herbal drugs (total extract or active fraction) and conventional drugs (single compound(s) drug)

1- Safety

2-Cost

3- Concentration and composition

4- Actions and uses

Characteristics of herbal Drugs

1- Safety

They are generally safer with fewer side effects.

2- Cost

1- They are generally cheaper than conventional medicines.

2- Nearly 1/3 of conventional drugs are still extracted from the opium poppy because it is more difficult and costly to synthesize them chemically Including morphine, codeine,

Characteristics of herbal Drugs

3- Concentration and composition.

a- They contain a wide variety of different com (active and inactive) (such as cellulose, starches and sugars).

b- Active compounds

1- Present in lower concentrations than in conventional drugs. So, many of them are used in the form of extract to concentrate the active compound(s).

2- Vary with many factors (amount of sunlight, rainfall, temperature, soil and season).



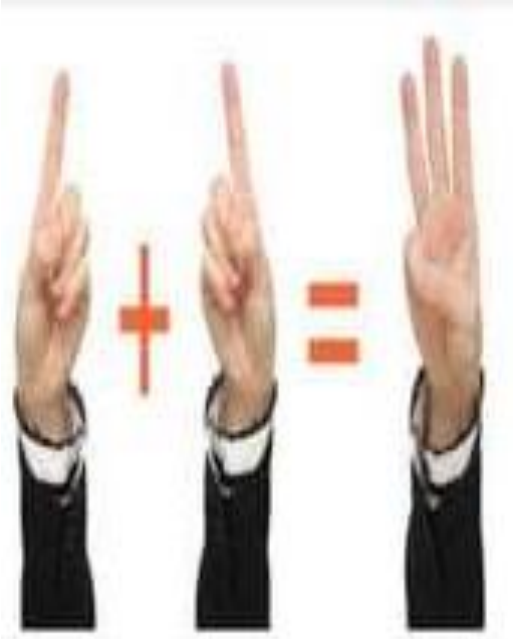
Characteristics of herbal Drugs

4- Actions and use:

due to complexity of the mixture of the herbal products it may exert multiplicity of uses and mechanisms.

1- Synergistic or antagonistic activities of the compounds, which determine their mode of use. Their power lies in synergistic interaction between all ingredients (known and unknown)

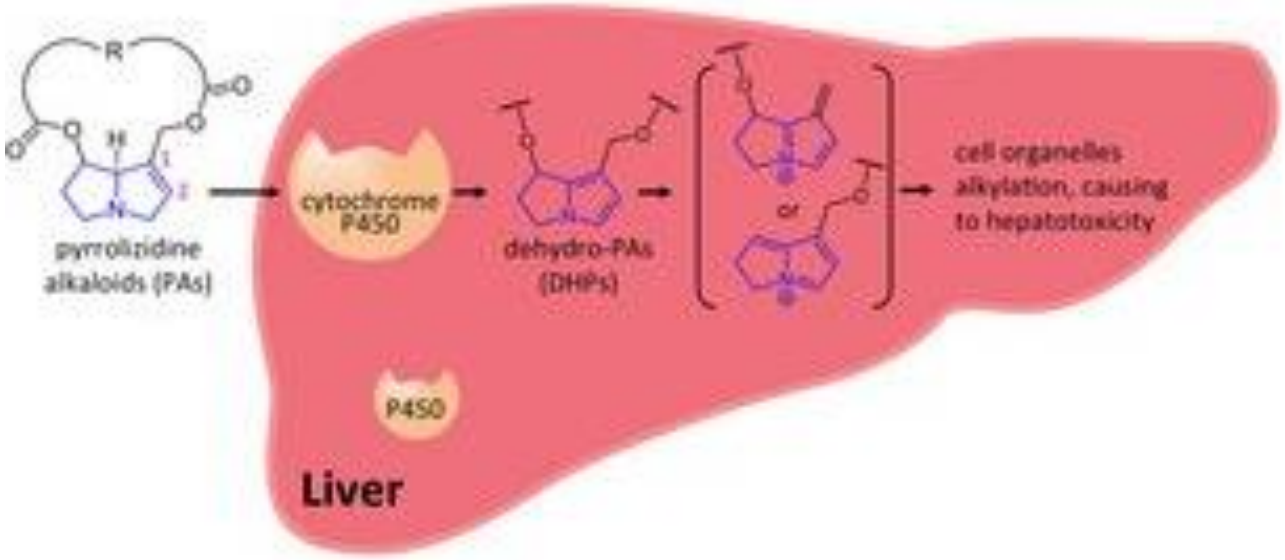
e.g. Rhubarb (anthraquinone & tannin)??.





Characteristics of herbal Drugs

2-Plants may also contain active and toxic compounds such as pyrrolizidine alkaloids e.g senecionine in Senecio which are converted in the liver into hepatotoxic and carcinogenic metabolites



Characteristics of herbal Drugs

3- They may support the general health through the prophylaxis against certain diseases and detoxification of body from free radicals and carcinogenic toxic compounds e.g. antioxidants, antihepatotoxics (protect against cancer and liver diseases), antihyperlipidemics against

4- Certain herbs are used as remedies and functional foods at the same time e.g. garlic, citrus, grapes, green and black tea.



Some commonly used Standardized herbal supplements.

Name	Standard content	Uses
1-Ginkgo (Ginkgo biloba)	Ginkgo flavones glycosides Total Ginkgolides (Lactones)	for Improving cerebral circulation and memory.
2-St. John's wort (Hypericum perforatum)	Naphthodianthrones (hypericin)	antidepressant
3-Saw palmetto	Sterols as β-sitosterol fatty acids	benign prostatic hyperplasia (BPH)
4-Valerian	Valerenic acid	sleep-inducing agent
5-Echinacea	Cichoric Acid Echinacoside	immunostimulant

How are herbs used?

- Herbal supplements are classified as dietary supplements by the U.S. Dietary Supplement Health and Education Act (DSHEA) of 1994.
- The FDA defines a dietary supplement as "...any product taken by mouth that contains a so-called 'dietary ingredient' and its label clearly states that it is a dietary supplement."
- Per the provisions of DSHEAI herbal supplements herbal supplements must be manufactured according to good manufacturing practices.

How are herbs used?

- Standardized herbal supplements are the best way to ensure proper dosages and effects similar to human clinical trials.
- Several herbs are often used together to enhance effectiveness and synergistic actions and to reduce toxicity.
- Health care providers must take many things into account when recommending herbs. For example, the species and variety of the plant, the plant's habitat, how it was stored and processed, and whether or not there are contaminants (including heavy metals and pesticides).

How do herbs work?

✓ For most herbs, the specific ingredient that causes a therapeutic effect is not known. Whole herbs contain many ingredients, and it is likely that they work together to produce the desired medicinal effect.

✓ Many factors determine how effective an herb will be. For example, the type of environment (climate, bugs, soil quality) in which a plant grew will affect its components, as will how and when it was harvested and processed.

Side effects &/or toxic reactions to herbal medicines

Rare as they are generally safe except minor allergic reactions They can arise from:

1- Mislabeling

Proper identification of the plant and labeling with common name and Latin name prevent this

2- Intensive addition of unnatural synthetic substances.

Chemical analysis of some Chinese herbal remedies used for arthritis proved that they contain some synthetic anti-inflammatory drugs such as indomethacin and phenylbutazone.

Thus, the toxic manifestations produced through the use of such products are due to synthetic additives

Side effects &/or toxic reactions to herbal medicines

3- Natural toxic contaminants

Plant material used for herbal medicines may be contaminated with micro- organisms or their by-products that can lead to serious adverse effects (aflatoxins).

4- Natural toxic ingredients

Herbal drugs containing pyrrolizidine alkaloids have been used since long times, but the first case reported about their hepatotoxicity appeared few years ago.

Contaminant Detection Methods

•Heavy Metals (Lead, Mercury, Arsenic,Cadmium):

•ICP-MS (Inductively Coupled Plasma Mass Spectrometry) detects heavy metal contamination.

•Detects lead, arsenic, and mercury in **ginseng, turmeric,**

•Microbial Contamination (Fungi, Bacteria, Aflatoxins):

•HPLC identifies fungal toxins (e.g., aflatoxins in herbal teas).

Contaminant Detection Methods

- **Adulteration with Pharmaceuticals:**
- Chromatographic techniques (HPLC, LC-MS) can detect synthetic drugs added illegally to herbal supplements.
- Example:
- Some Chinese arthritis remedies contained ibuprofen, steroids, and NSAIDs without disclosure.

Examples for Undesirable Effects of Herbal Medicines

Excessive ingestion:

Ginseng:

- Excessive doses of Ginseng have been reported to cause agitation, insomnia, and raised blood pressure and have been referred to as abuse of the remedy.
- However, side-effects have also been reported for Ginseng following the ingestion of recommended doses.



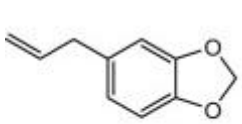
Examples for Undesirable Effects of Herbal Medicines

Liquorice:

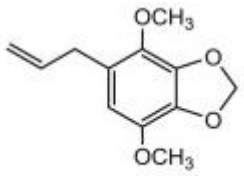
- Excessive ingestion of Licorice has resulted in typical side effects of corticosteroid type e.g; edema as a result of strong water retention and rise in blood pressure.

Parsley:

- Parsley volatile oil contains apiole which structurally related to the recognized hepato-carcinogen, safrole.



safrole



apiol

- Ingestion of apiole has resulted in a number of cases of fatal poisoning.

Examples for Undesirable Effects of Herbal Medicines

Kava kava:

Traditionally used for anxiety but linked to hepatotoxicity in high doses.

Several European countries banned kava due to reports of liver failure.

Ephedra:

Previously used in weight loss supplements but caused hypertension, stroke, and death.

Banned by FDA in 2004 after multiple adverse event reports.

Examples for Undesirable Effects of Herbal Medicines

Hypersensitivity reactions:

Chamomile:

- sesquiterpene lactones as matricin possess allergic properties.

Phototoxic reactions:

Parsley:

- Furanocoumarins as psoralens, compounds known to cause phototoxic reactions, are constituents of Parsley.
- Excessive ingestion of Parsley has been associated with the development of photosensitive rashes which are resolved once Parsley consumption ceased.

Examples for Undesirable Effects of Herbal Medicines

Star anise

- Example of Herbal Tea Toxic Alert by FDA against the drinking of teas brewed from star anise.
- In fact, medical conditions associated with star anise teas include seizures, vomiting and rapid eye movement.-
- One problem is that there is more than one type of star anise.
- The Chinese star anise is considered safe.
- A closely related species, the Japanese star anise, contains sikimitoxin and is toxic

Examples for Undesirable Effects of Herbal Medicines



- Once star anise has been dried and processed, it is not possible to visually distinguish between the Chinese and Japanese forms. Some teas may actually be Japanese or a mixture of the two types.
- Until the FDA has "analyzed" the situation, it recommends that the consumer avoid all teas containing star anise.

Herb-Drug Interactions

- Some herbs interact with prescription medications, leading to reduced drug efficacy or increased toxicity.

Herb	Interaction
St. John's Wort	Induces CYP3A4 → Reduces effectiveness of oral contraceptives, antidepressants, and immunosuppressants
Ginkgo biloba	Increases bleeding risk when taken with anticoagulants (warfarin, aspirin)
Licorice (Glycyrrhiza glabra)	Can cause hypertension and hypokalemia if used with diuretics or corticosteroids
Ginseng	May lower blood sugar → Risky for diabetics on hypoglycemic medications

Regulatory Aspects in Phytotherapy

- Phytotherapy is regulated differently across the world, with varying levels of oversight and classification. Below are key international regulatory frameworks:
- **1-World Health Organization (WHO):**
- WHO has established guidelines on the quality, safety, and efficacy of herbal medicines.
- It encourages good agricultural and collection practices (GACP) to ensure the proper cultivation, harvesting, and processing of medicinal plants.
- WHO monographs provide scientific validation and standards for commonly used medicinal plants.

Regulatory Aspects in Phytotherapy

2-European Medicines Agency (EMA):

- The Committee on Herbal Medicinal Products (HMPC) oversees herbal medicines in the EU.
- Traditional Herbal Medicinal Products Directive (THMPD) 2004/24/EC simplifies the approval process for herbal medicines used for at least 30 years (including 15 years within the EU).
- Requires standardization, quality control, and proof of traditional use but not necessarily clinical trials.

Regulatory Aspects in Phytotherapy

3-United States Food and Drug Administration (FDA):

- Herbal products are regulated as dietary supplements under the Dietary Supplement Health and Education Act (DSHEA) of 1994.
- Unlike conventional drugs, herbal supplements do not require pre-market approval but must follow good manufacturing practices (GMP) and truth-in-labeling standards.
- FDA can ban products proven to be unsafe (e.g., ephedra).

Clinical Evidence and Modern Research in Phytotherapy

- Phytotherapy is moving towards evidence-based medicine through rigorous clinical research.
- **Clinical Trials:**
- Well-conducted randomized controlled trials (RCTs) validate herbal medicines (e.g., ginkgo biloba for cognitive function, turmeric for inflammation).

Challenges in Herbal Clinical Trials:

1-Variability in herbal extracts:

- Unlike synthetic drugs, plant constituents vary due to climate, soil, and harvesting methods.

2-Standardization difficulties:

- Establishing a consistent dose of active ingredients is challenging.

3-Placebo effect:

- Herbal medicines often have strong placebo effects, making efficacy assessment complex.

Phytotherapy vs. Traditional Herbal Medicine

Aspect	Phytotherapy	Traditional Herbal Medicine
Definition	Evidence-based use of medicinal plants, often in standardized extracts	Traditional use of plants for healing, often based on cultural practices
Scientific Basis	Focuses on phytochemistry, pharmacology, and clinical trials	Based on historical knowledge, folklore, and holistic principles
Regulation	Subject to regulations (e.g., EMA, FDA)	Often unregulated or regulated under traditional medicine frameworks
Standardization	Extracts are standardized for active compounds	Whole plants or crude preparations with variable potency
Safety & Toxicity	Assessed through clinical studies and toxicological evaluations	May lack safety assessments, risk of contaminants or interactions

Challenges in Phytotherapy

1-Standardization Issues:

- Herbal products may vary in potency due to factors like soil, climate, and storage.
- Lack of global standardization protocols for ensuring consistent active ingredient levels.

2-Adulteration and Contamination:

- Some herbal products are adulterated with pharmaceuticals (e.g., synthetic steroids in weight loss supplements).
- Contamination risks include heavy metals, pesticides, and microbial growth (e.g., aflatoxins in herbal teas).

Challenges in Phytotherapy

3-Regulatory and Legal Barriers:

- Different countries have inconsistent regulations, making international trade and clinical trials difficult.
- Some herbs are banned in certain regions due to safety concerns (e.g., kava banned in several European countries due to hepatotoxicity

Emerging Trends in Phytotherapy

- **AI-Assisted Phytochemical Analysis:**

•Machine learning models predict biological activities of plant compounds.

•AI aids in drug discovery by analyzing vast phytochemical databases.

- **Personalized Phytotherapy:**

•Based on an individual’s genetics, microbiome, and metabolism, herbal treatments can be tailored for better efficacy.

•Example: Turmeric metabolism varies among individuals due to genetic differences in liver enzymes.